

SIMATS ENGINEERING

CO5- ASSESSMENT TOOL 3 – Technical Presentation

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

COURSE OUTCOME COVERED

CO5: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications. (BTL 5)*

Assessment Tool Weightage: 10%

Time Duration: 1 Hour

QUESTIONS: 5

Total Marks:50

Code of Conduct:

I _Dhanush_ (192512132_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

G.Dhanush

1. System Analysis Presentation

Present a reverse engineering analysis of an AI-based system, explaining its architecture, components, and working process using diagrams.

2. Architecture Reconstruction Demo

Prepare a technical presentation to reconstruct the architecture of a given software/AI system, highlighting module interactions and data flow.

3. Algorithm & Logic Identification

Present how you identified algorithms and decision-making logic in a reverse-engineered system, including methodology and tools used.

4. Data Flow & Processing Pipeline

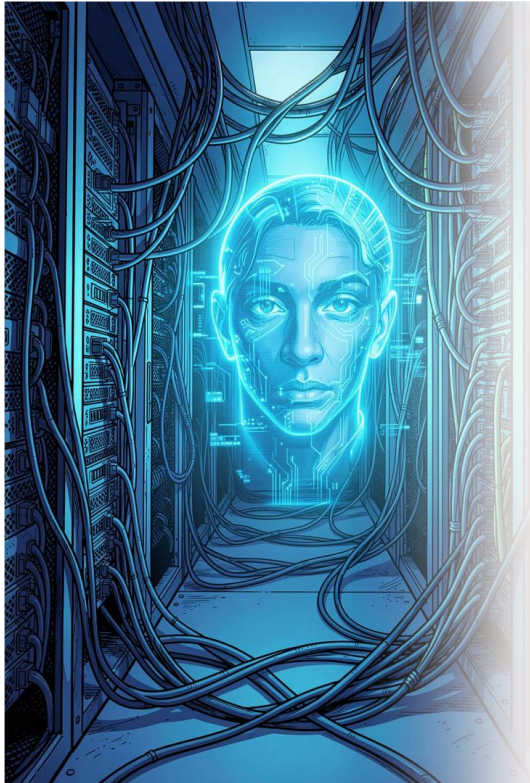
Explain through a presentation the data flow, processing pipeline, and communication mechanisms in a reverse-engineered distributed system.

5. Enhancement & Redesign Proposal

Deliver a technical presentation proposing improvements or redesign strategies for an existing system based on reverse engineering findings.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness



CSA 1709 ARTIFICIAL INTELLIGENCE

Reverse Engineering an AI System—Technical Briefing

Presented By:
G.Dhanush
(192512132)

Guided By:
Dr.Rajaram.P

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Agenda

1 Objective & Scope

Define targets, assets, and limits for reverse engineering.

2 Architecture Reconstruction

Component mapping, interfaces, and deployment topology.

3 Algorithm Discovery

Techniques used to identify core models, heuristics, and decision logic.

4 Data Pipeline Analysis

Ingress, preprocessing, feature engineering, and communications.

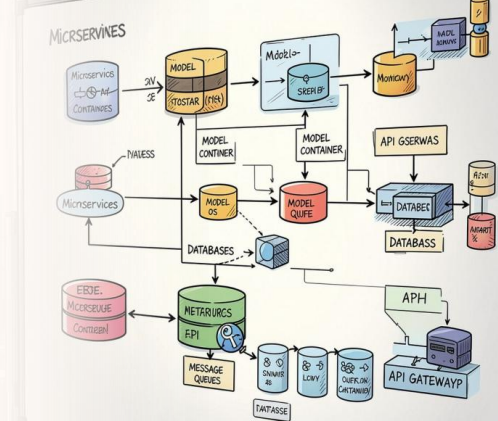
5 Recommendations

Security, scalability, and maintainability improvements.

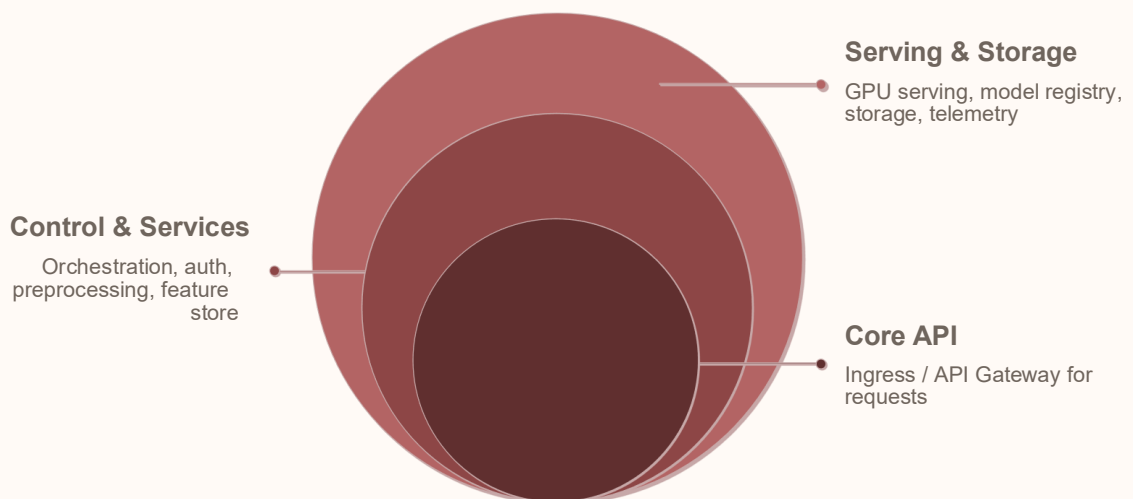
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How We Reconstructed the System

Approach: iterative artifact collection → dynamic observation → static analysis → behavioural inference. Inputs: binaries, container images, network traces, configuration files, and test harnesses. Goal: reproducible architecture diagram and threat surface map



Architecture—High-level Reconstruction



Key insight: separation of control plane (orchestration, model registry) from data plane (stream processing, inference), with

asynchronous event bus enabling horizontal scaling.



Component Interaction & Communication

Primary channels: REST/gRPC for synchronous control, Kafka -like pub/sub for event streams, object storage for large artefacts.

Authentication: token -based (JWT) observed; TTL and key rotation inconsistent across services.

- Service-to-service auth: mutual TLS observed on some paths, absent on others
- Model updates: push via CI pipeline into container registry and model registry
- Monitoring: Prometheus metrics, ELK -stack logs aggregated centrally

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Identifying Algorithms & Decision Logic

Methodology: behavior -driven probing, input -output correlation, model fingerprinting, static code signatures, and feature -impact analysis. Tools: dynamic instrumentation, model distillation, differential testing, decompilers, and explainability pr obes (SHAP/LIME).



Behavioral Probing

Systematic input perturbation to infer model boundaries, thresholds and fallback rules.



Static & Binary Analysis

Reverse engineer model IO, preprocess pipelines, activation functions and custom layers.



Explainability Tools

Feature attribution to map decision logic back to input signals.

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Observations

- Inconsistent auth between services — lateral movement risk.
- Feature store lacks strong RBAC and encryption -at-rest in parts.
- Model poisoning exposure: unvalidated training inputs in active learning loop.
- Telemetry gaps: some inference nodes not shipping logs, impeding detection.



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Enhancement & Redesign Proposal—Technical Priorities

Strong Service Authentication

Mutual TLS everywhere + centralized short-lived certificates and automated rotation.

Secure Feature Store

Encrypt -at-rest, column-level access controls, and immutable lineage metadata.

Model Governance

Signed model artefacts, deterministic CI promotion, and canary testing for inference drift.

Observability & Response

End-to-end telemetry, anomaly detection on inputs, and automated incident playbooks.

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